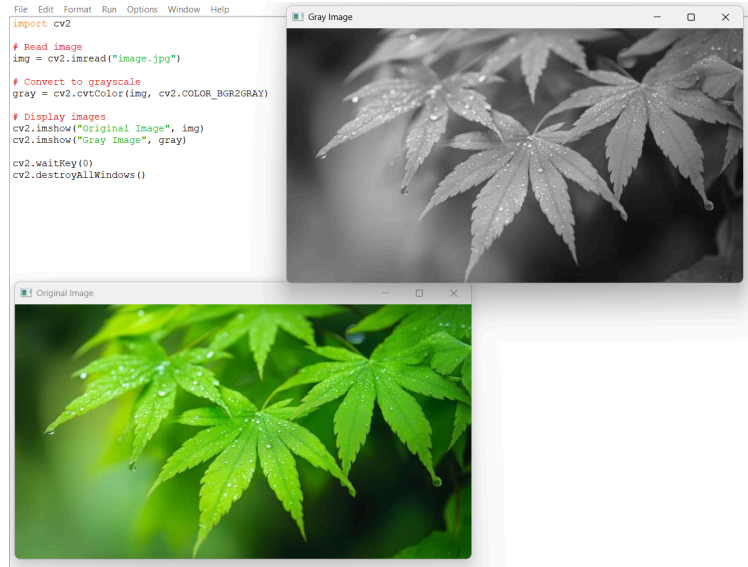


LAB EXPERIMENTS

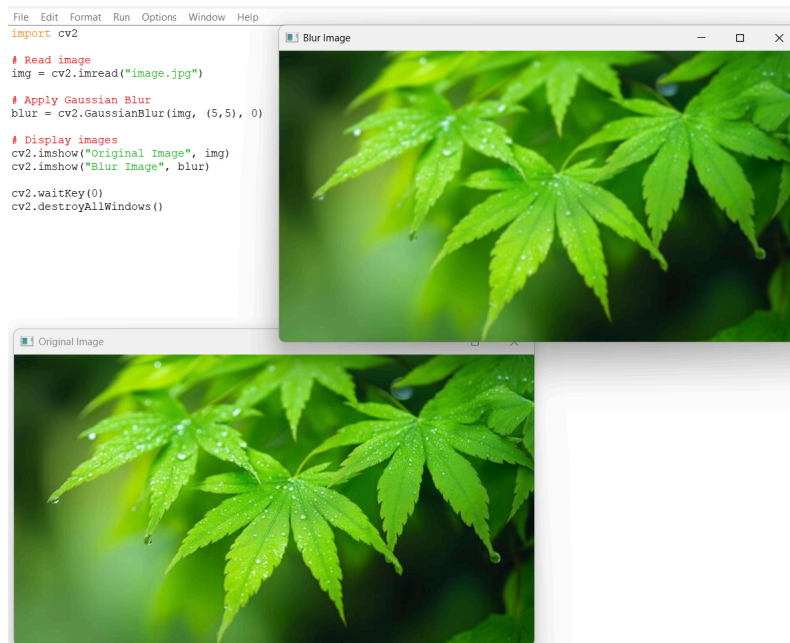
NAME: KAVIYA V
COURSE CODE: ITA0512

REG NO: 192421022
COURSE NAME: COMPUTER VISION

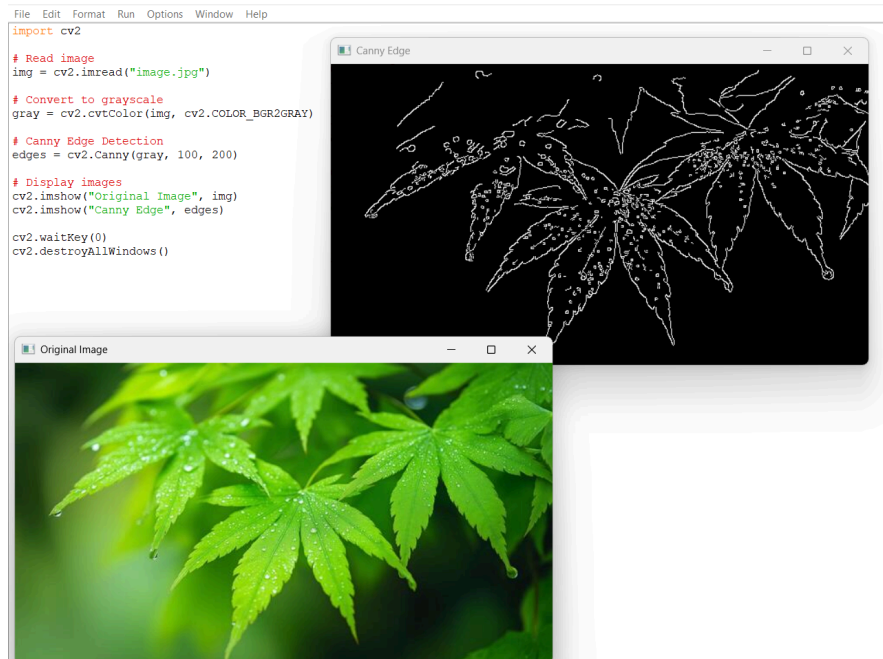
1. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to Gray-scale.



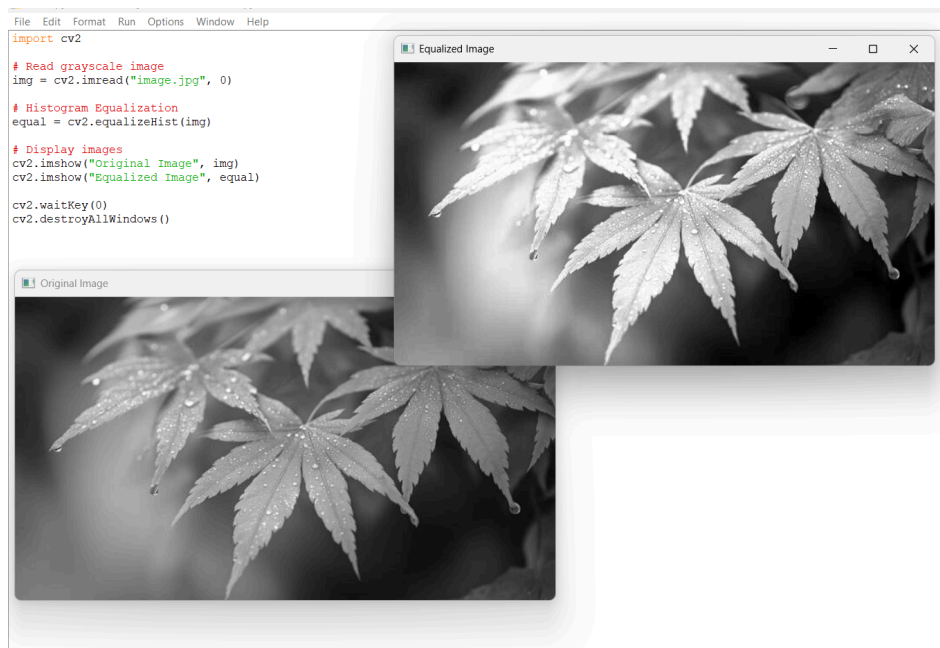
2. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to Blur using Gaussian Blur



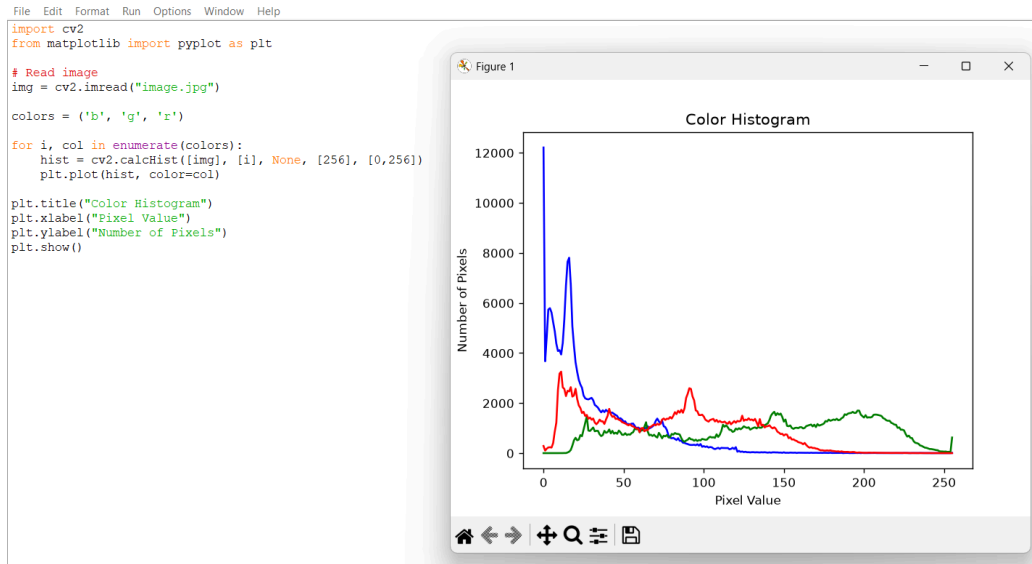
3. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to show outline using Canny function.



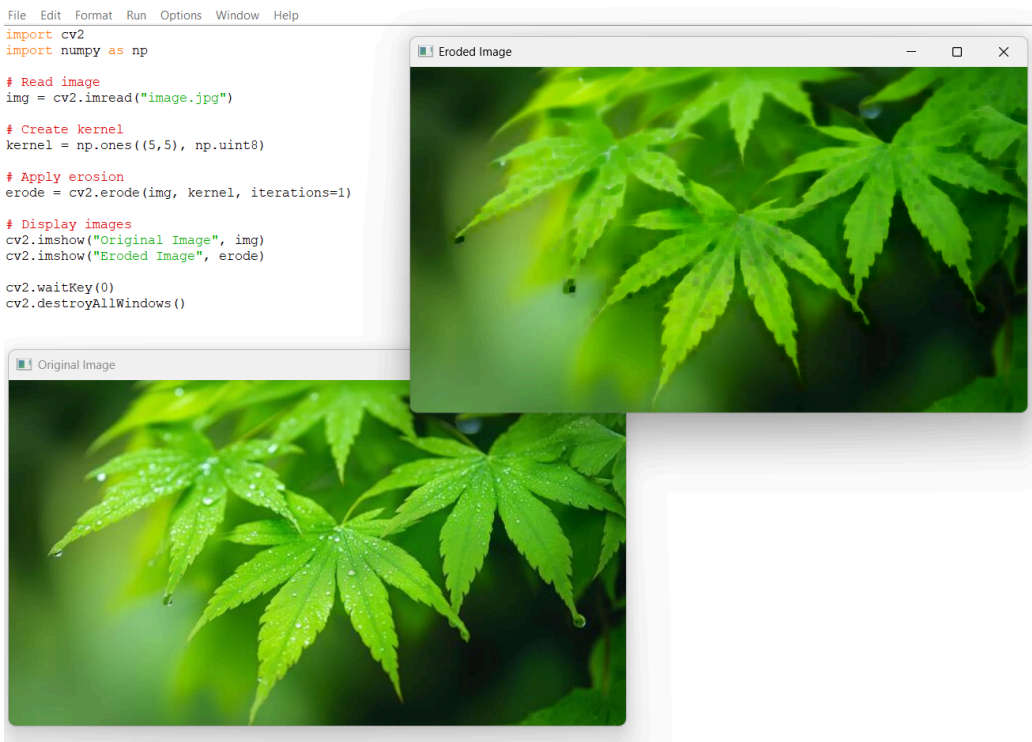
4. Implement histogram equalization on the given image and compare it with the original image using Open CV.



5. Write a Python function to analyze the histogram of the given input image based on color levels using Open CV.



6. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to erode using Erode function.



7. Perform basic video processing operations on the captured video. Read captured video in python and display the video, in slow motion and in fast motion.

```
import cv2

# Read video
cap = cv2.VideoCapture(r"C:\Users\Kaviya V\Downloads\video.webm")

if not cap.isOpened():
    print("Cannot open video")
    exit()

print("Choose Playback Speed")
print("1 - Slow Motion")
print("2 - Normal")
print("3 - Fast Motion")

choice = input("Enter your choice: ")

if choice == "1":
    delay = 100 # Slow Motion
elif choice == "3":
    delay = 10 # Fast Motion
else:
    delay = 30 # Normal Speed

while True:
    ret, frame = cap.read()
    if not ret:
        break

    cv2.imshow("Video Playback", frame)

    if cv2.waitKey(delay) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

8. Perform basic Image Handling and processing operations on the image is to read an image in python and Dilate an Image using Dilate function

```
import cv2
import numpy as np

# Read image
img = cv2.imread("image.jpg")

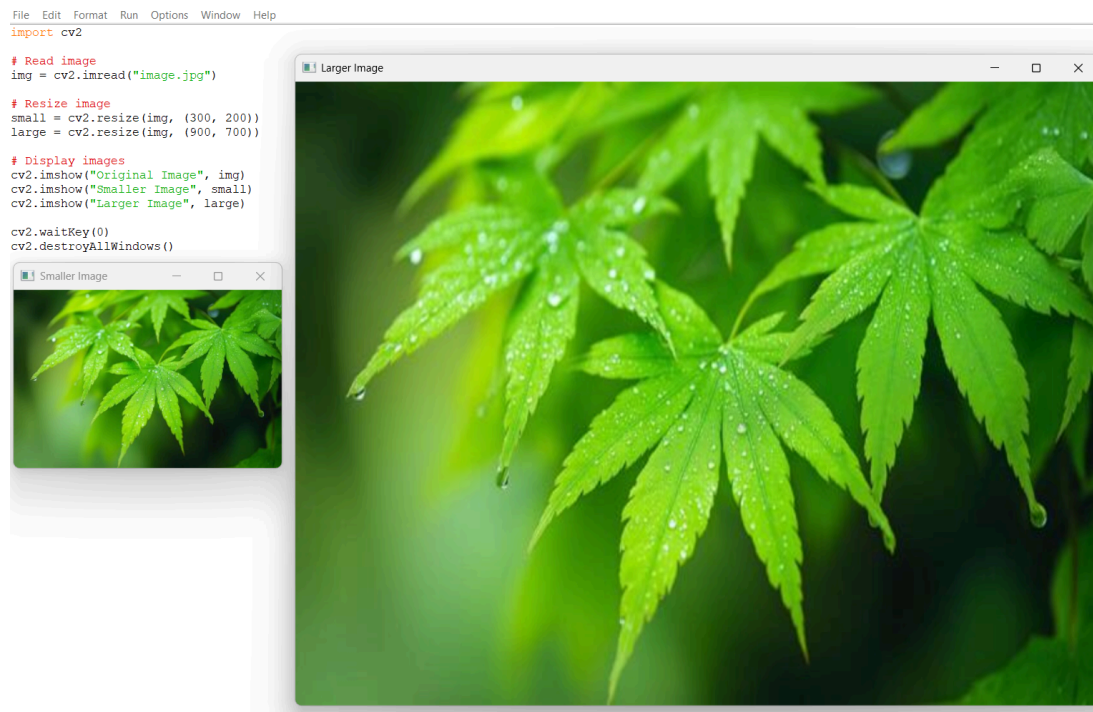
# Create kernel
kernel = np.ones((5,5), np.uint8)

# Apply dilation
dilate = cv2.dilate(img, kernel, iterations=1)

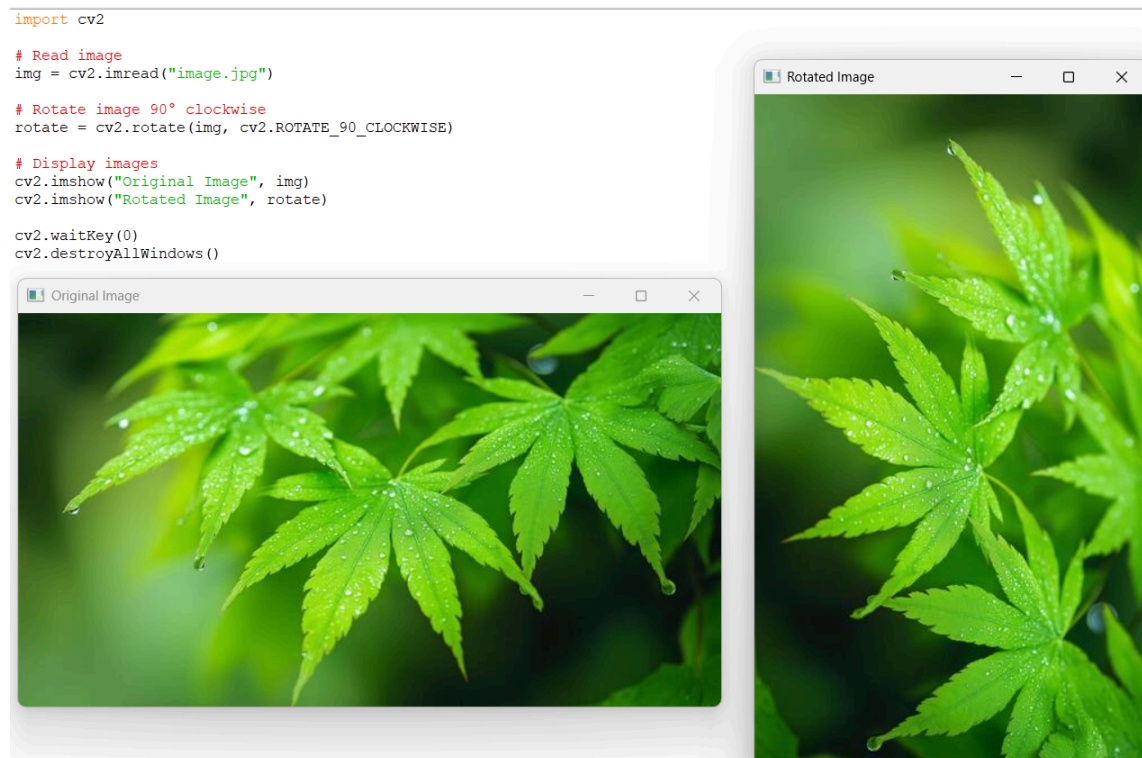
# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Dilated Image", dilate)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

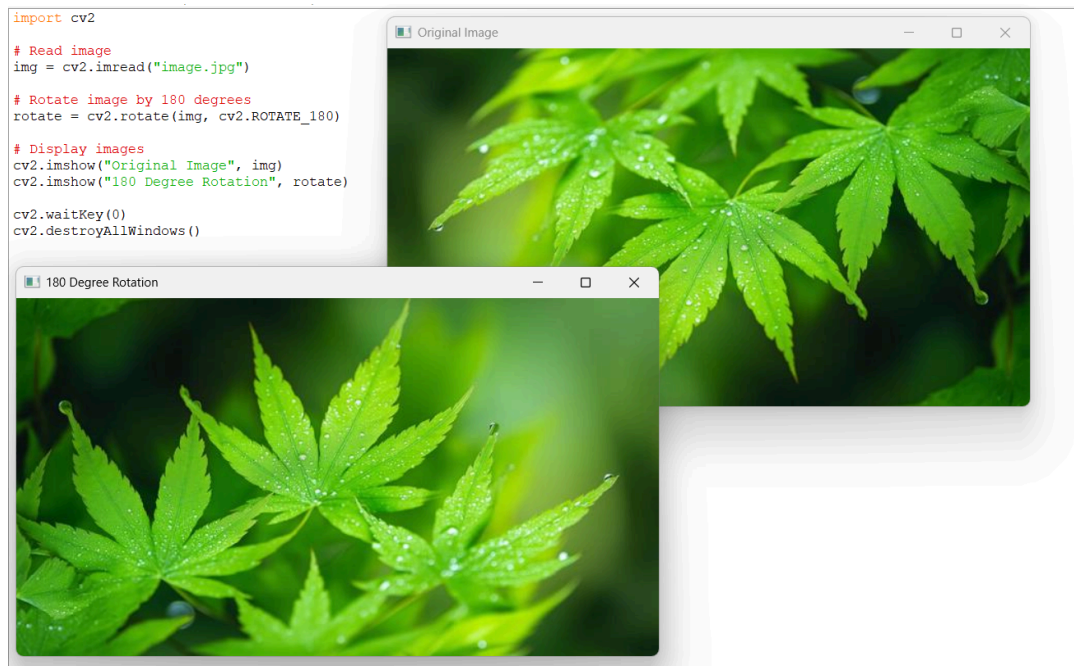

9. Implement the image scaling techniques to resize the images to both bigger and small sizes



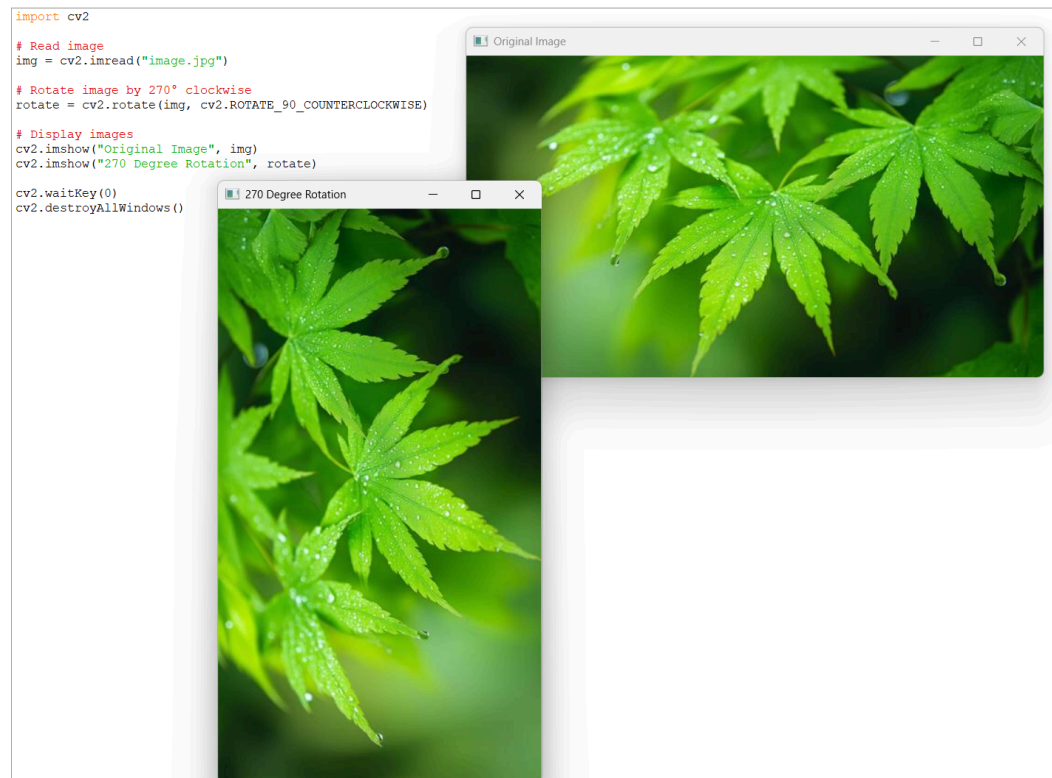
10. Perform a 90-degree rotation clockwise along the y-axis for the given image.



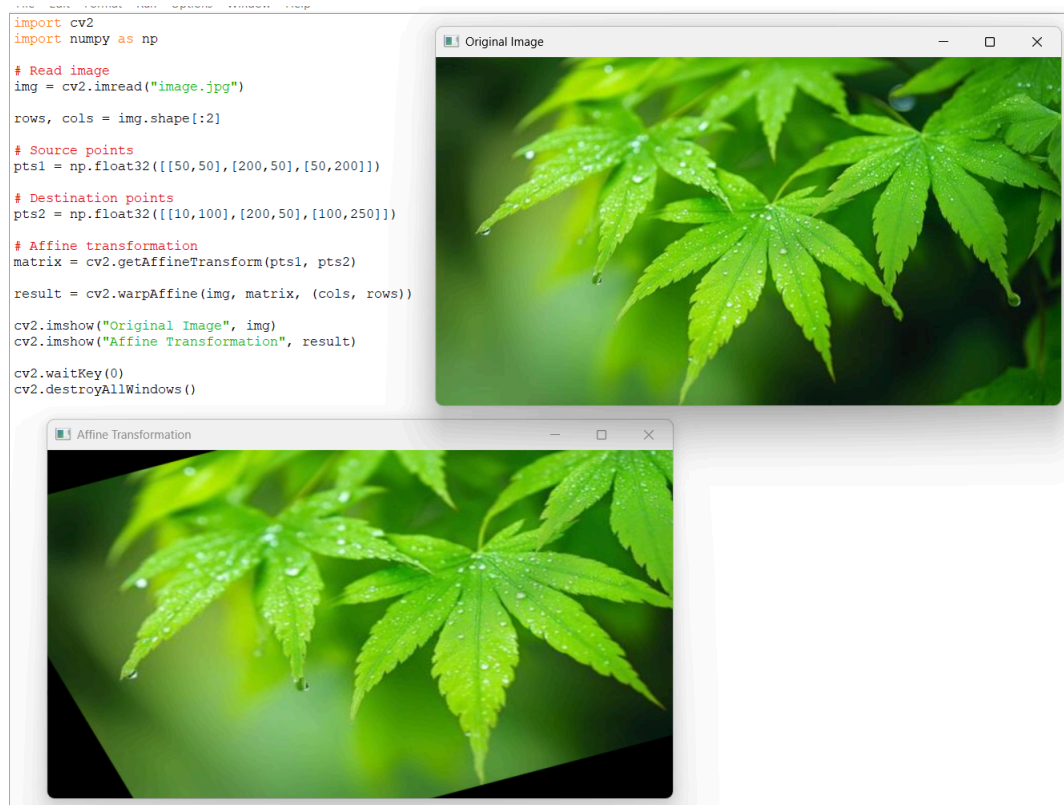
11 .Perform a 180-degree rotation clockwise along the y-axis for the given image.



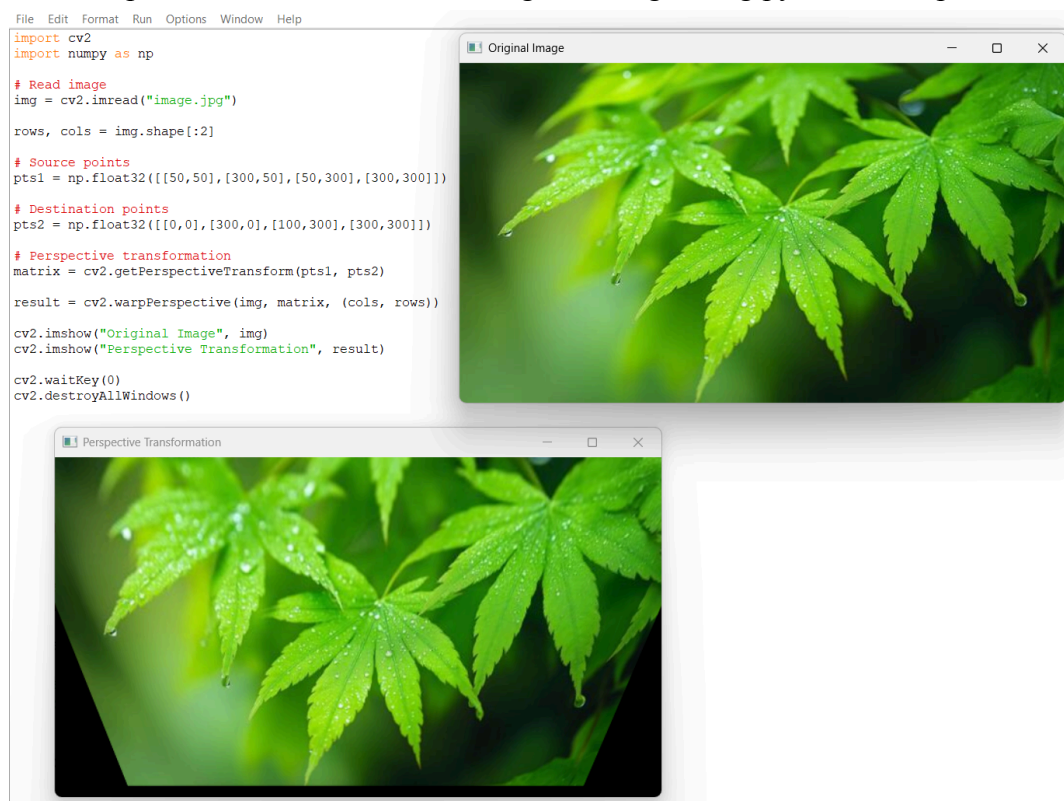
12.Perform a 270-degree rotation clockwise along the y-axis for the given image.



13. Perform Affine Transformation on the given image using python and Open CV



14. Perform Perspective Transformation on the given image using python and Open CV.



15. Perform basic Image Handling and processing operations on the image is to read an image in python and detect the corners in the image using Harris Corner Detection function

```
File Edit Format Run Options Window Help
import cv2
import numpy as np

# Read image
img = cv2.imread("image.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
gray = np.float32(gray)

# Harris Corner Detection
corners = cv2.cornerHarris(gray, 2, 3, 0.04)

# Mark corners
img[corners > 0.01 * corners.max()] = [0, 0, 255]

cv2.imshow("Harris Corner Detection", img)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

